

sensors etc. The machine needs to know about its surrounding in order to perform an action.

- **Movement** – Either the machine should be capable of movement in its surrounding environment or it should have moving parts that somehow interact with the environment. This part of the robot is called ‘actuators’.
- **Energy** – The machine should have a power source that provides it energy. It can be anything from solar power to rechargeable-batteries.
- **Intelligence** – The most defining characteristic of a robot, intelligence is what primarily sets apart a robot from a machine. It should possess some sort of intelligence such that it processes and takes decisions about its action based on the data it collects from its sensors.

Unlike popular misconception, a robot doesn’t have to necessarily look like us humans. Infact, most of today’s robots don’t. An android robot that looks like us and is capable of mimicking our actions and features is still some time away.

History and origin of the word ‘robot’

The word ‘robot’ is old and didn’t always mean what it does in this age. Robot meant ‘forced labor’ and was used to refer to the system of slavery spread across central Europe, until it was abolished in the year 1848.

The word again popped up in the year 1920, when Karel Capek, a Czech writer, wrote a play titled *Rossum’s Universal Robots*. In the play, workers were mass produced from synthetic organic material and were called robots.

Strangely, the play was not only the first instance of robots but also the first popular media to showcase a robot uprising and a dystopian futuristic world. Karel Capek didn’t know that his play was going to give birth to a whole new sub-genre of sci-fi where robots take over the world and turn against humanity. I guess we have someone to thank for the Terminator franchise.



Not every robot looks like a humanoid and has arms and legs. The Roomba vacuum cleaner is a robot that cleans your house’s floors on its own.